LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control horizontal and vertical movement, control of RPM, torque reaction, tail rotor compensations, contra rotating rotors.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control horizontal and vertical movement, control of RPM, torque reaction, tail rotor compensations, contra rotating rotors.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight control stick positions and cyclic/collective inputs on simulator or ground station; Instructor supervision required for safety.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the cyclic pitch control?
(b) How does the collective pitch control affect the main rotor?
(c) What is the immediate effect of torque reaction on the helicopter fuselage?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Cyclic control for horizontal movement • Collective control for vertical movement • RPM control via collective and throttle • Torque reaction and anti-torque systems • Tail rotor compensation mechanisms	L & Ds	PPT, IB	15 min
(b)	• Contra-rotating rotor configurations • Advantages of contra-rotating systems • Elimination of tail rotor requirements • Efficiency gains in power transmission • Comparison with single main rotor designs	L & Ds	PPT, PDF	20 min
(c)	• Vibration checks during flight • Adjustments for rotor balance • Impact of torque on stability • RPM fluctuation management • Tail rotor effectiveness verification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the tail rotor compensates for torque reaction.
(b) Describe the role of contra-rotating rotors in eliminating torque effects.
(c) What steps are taken to check for vibration during flight?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between single main rotor and contra-rotating rotor systems, referencing LP 02 concepts on rotor dynamics.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Main rotor movements, dy-symmetry of lift feathering, flapping, dragging, hook's joint effect, flapping to equality, phase lag and advance angle concepts	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min